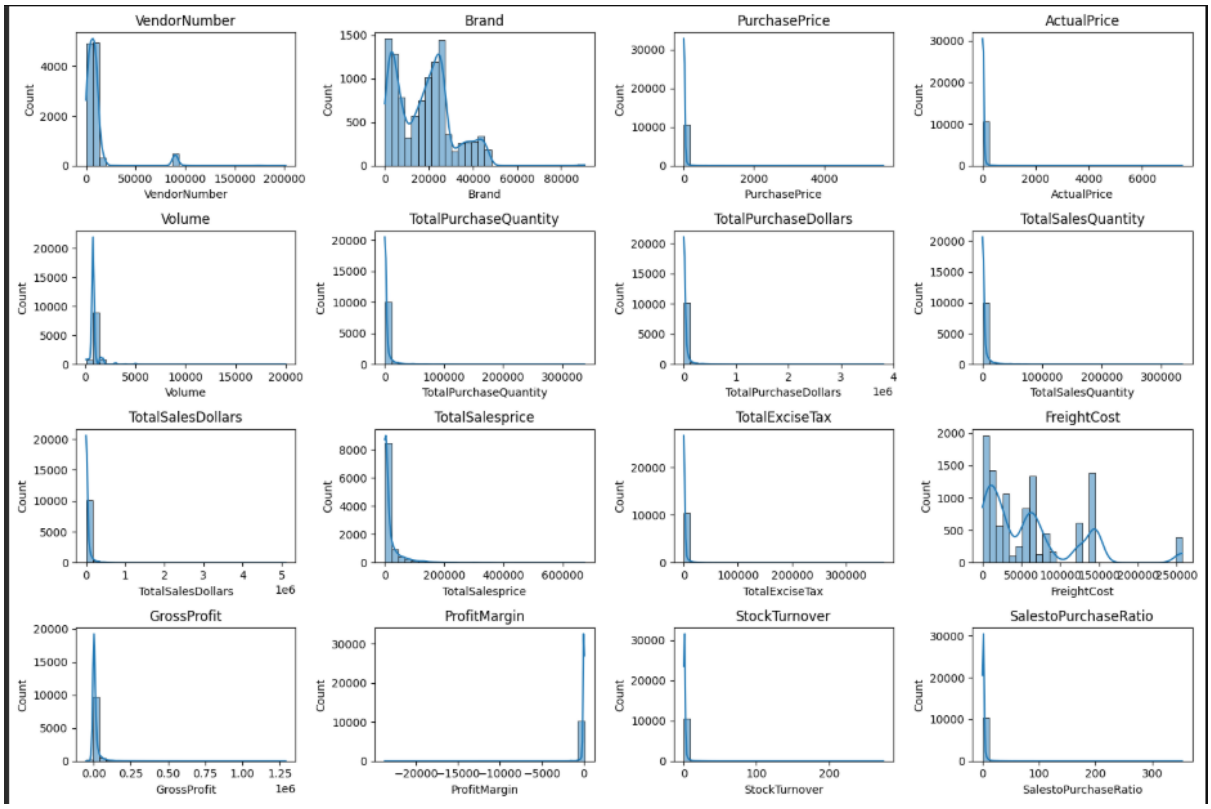
Exploratory Data Analysis Insights

Summary Statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	10650.649458	18753.519148	2.000000	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	18039.228769	12662.187074	58.000000	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	24.385303	109.269375	0.360000	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	35.643671	148.246016	0.490000	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	847.360550	664.309212	50.000000	750.000000	750.000000	750.000000	2.000000e+04
TotalPurchaseQuantity	10692.0	3140.886831	11095.086769	1.000000	36.000000	262.000000	1975.750000	3.376600e+05
TotalPurchaseDollars	10692.0	30106.693372	123067.799627	0.710000	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesQuantity	10692.0	3077.482136	10952.851391	0.000000	33.000000	261.000000	1929.250000	3.349390e+05
TotalSalesDollars	10692.0	42239.074419	167655.265984	0.000000	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesprice	10692.0	18793.783627	44952.773386	0.000000	289.710000	2857.800000	16059.562500	6.728193e+05
TotalExciseTax	10692.0	1774.226259	10975.582240	0.000000	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	61433.763214	60938.458032	0.090000	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	12132.381048	46224.337964	-52002.780000	52.920000	1399.640000	8660.200000	1.290668e+06
ProfitMargin	10514.0	-15.885227	447.289882	-23730.638953	15.353839	30.778375	40.210967	9.971666e+01
StockTurnover	10692.0	1.706793	6.020460	0.000000	0.807229	0.981529	1.039342	2.745000e+02
SalestoPurchaseRatio	10692.0	2.504390	8.459067	0.000000	1.153729	1.436894	1.665449	3.529286e+02



Negative & Zero Values:

Gross Profit: Minimum of 52,002.78, indicating potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.

Profit Margin: Has a minimum of ∞, which suggests instances where revenue is zero or even lower than the total cost, leading to extreme negative profit margins.

Total Sales Quantity & Sales Dollars: Some products show zero sales, indicating they were purchased but never sold. These may be slow-moving or obsolete stock, leading to inventory inefficiencies.

Outliers Detected by High Standard Deviations:

Purchase & Actual Prices: The maximum values (5,681.81 & 7,499.99) are significantly higher than the mean (24.39 & 35.64), indicating premium product offerings.

Freight Cost: Extreme variation from 0.09 to 257,032.07 suggests logistics inefficiencies, bulk shipments, or erratic shipping costs across different products.

Stock Turnover: Ranges from 0 to 274.5, suggesting some products sell rapidly while others remain unsold for long periods. A value greater than 1 indicates that sales for a product exceed the purchased quantity due to older stock fulfilling orders.

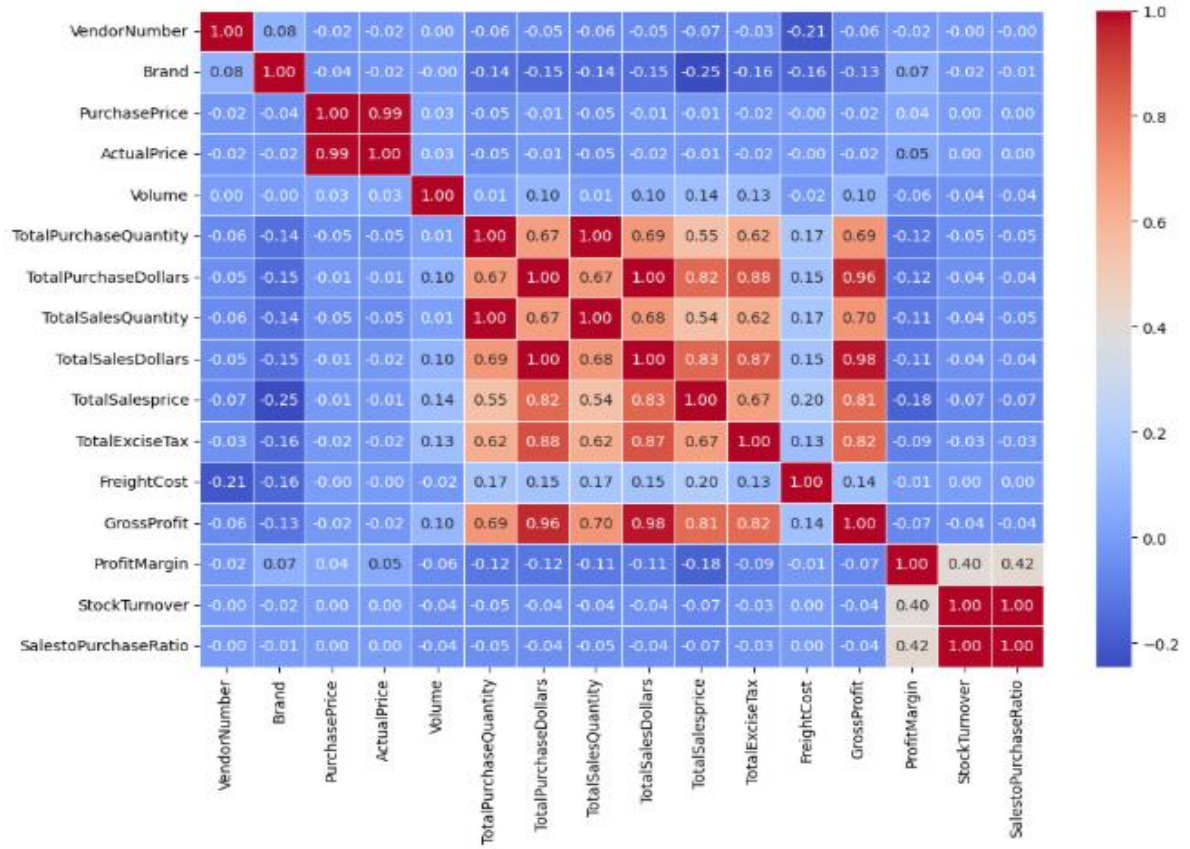
Data Filtering

To enhance the reliability of the insights, we removed inconsistent data points where:

- Gross Profit ≤0 (to exclude transactions leading to losses).

- Profit Margins 0 (to ensure analysis focuses on profitable transactions).
- Total Sales Quantity = 0 (to eliminate inventory that was never sold).

Correlation Insights



Purchase Price vs. Total Sales Dollars & Gross Profit: Weak correlation (-0.012 and -0.016), indicating that price variations do not significantly impact sales revenue or profit.

Total Purchase Quantity vs. Total Sales Quantity: Strong correlation (0.999), confirming efficient inventory turnover.

Profit Margin vs. Total Sales Price: Negative correlation (-0.179), suggesting increasing sales prices may lead to reduced margins, possibly due to competitive pricing pressures.

Stock Turnover vs. Gross Profit & Profit Margin: Weak negative correlation (-0.038 & -0.055), indicating that faster stock turnover does not necessarily equate to higher profitability.

Research Questions & Key Findings

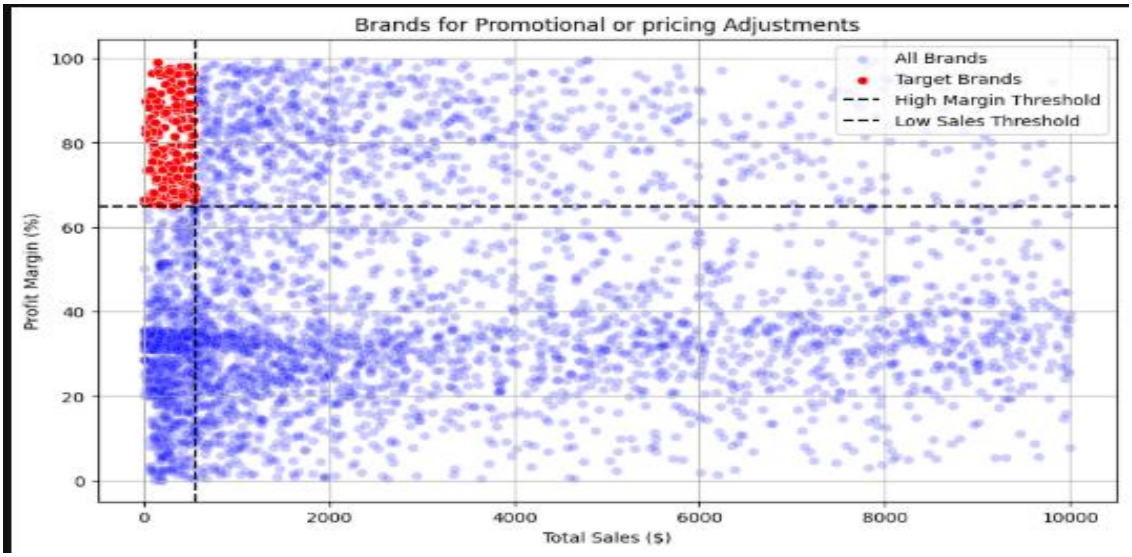
1. Brands for Promotional or Pricing Adjustments

Brand with low sales but high profit margins:

	Description	TotalSalesDollars	ProfitMargin
6199	Santa Rita Organic Svgn Bl	9.99	66.466466
2369	Debauchery Pnt Nr	11.58	65.975820
2070	Concannon Glen Ellen Wh Zin	15.95	83.448276
2188	Crown Royal Apple	27.86	89.806174
6237	Sauza Sprklg Wild Berry Marg	27.96	82.153076
...	...	...	...
5074	Nanbu Bijin Southern Beauty	535.68	76.747312
2271	Dad's Hat Rye Whiskey	538.89	81.851584
57	A Bichot Clos Marechaudes	539.94	67.740860
6245	Sbragia Home Ranch Merlot	549.75	66.444748
3326	Goulee Cos d'Estournel 10	558.87	69.434752

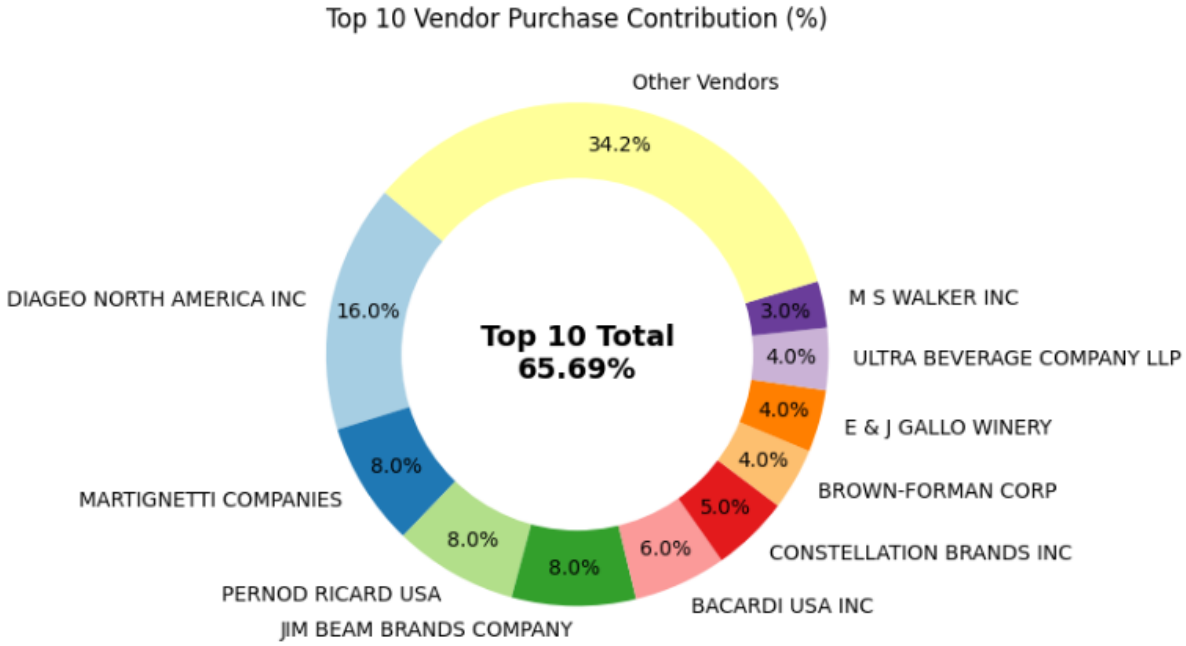
198 rows × 3 columns

198 brands exhibit lower sales but higher profit margins, which could benefit from targeted marketing, promotions, or price optimizations to increase volume without compromising profitability.



2. Top Vendors by Sales & Purchase Contribution

The top 10 vendors contribute 65.69% of total purchases, while the remaining vendors contribute only 34.31%. This over-reliance on a few vendors may introduce risks such as supply chain disruptions, indicating a need for diversification.



3. Impact of Bulk Purchasing on Cost Savings

Vendors buying in large quantities receive a 72% lower unit cost ($10.78 per unit vs. higher unit costs in smaller orders).

Bulk pricing strategies encourage larger orders, increasing total sales while maintaining profitability.

OrderSize	UnitPurchasePrice
Small	39.057543
Medium	15.486414
large	10.777625

4. Identifying Vendors with Low Inventory Turnover

Total Unsold Inventory Capital: $2.71M

Slow-moving inventory increases storage costs, reduces cash flow efficiency, and affects overall profitability.

Identifying vendors with low inventory turnover enables better stock management, minimizing financial strain.

VendorName	StockTurnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

	VendorName	UnsoldInventoryValue
25	DIAGEO NORTH AMERICA INC	722.21K
46	JIM BEAM BRANDS COMPANY	554.67K
68	PERNOD RICARD USA	470.63K
116	WILLIAM GRANT & SONS INC	401.96K
30	E & J GALLO WINERY	228.28K
79	SAZERAC CO INC	198.44K
11	BROWN-FORMAN CORP	177.73K
20	CONSTELLATION BRANDS INC	133.62K
61	MOET HENNESSY USA INC	126.48K
77	REMY COINTREAU USA INC	118.60K

5. **Profit Margin Comparison: High vs. Low-Performing Vendors**

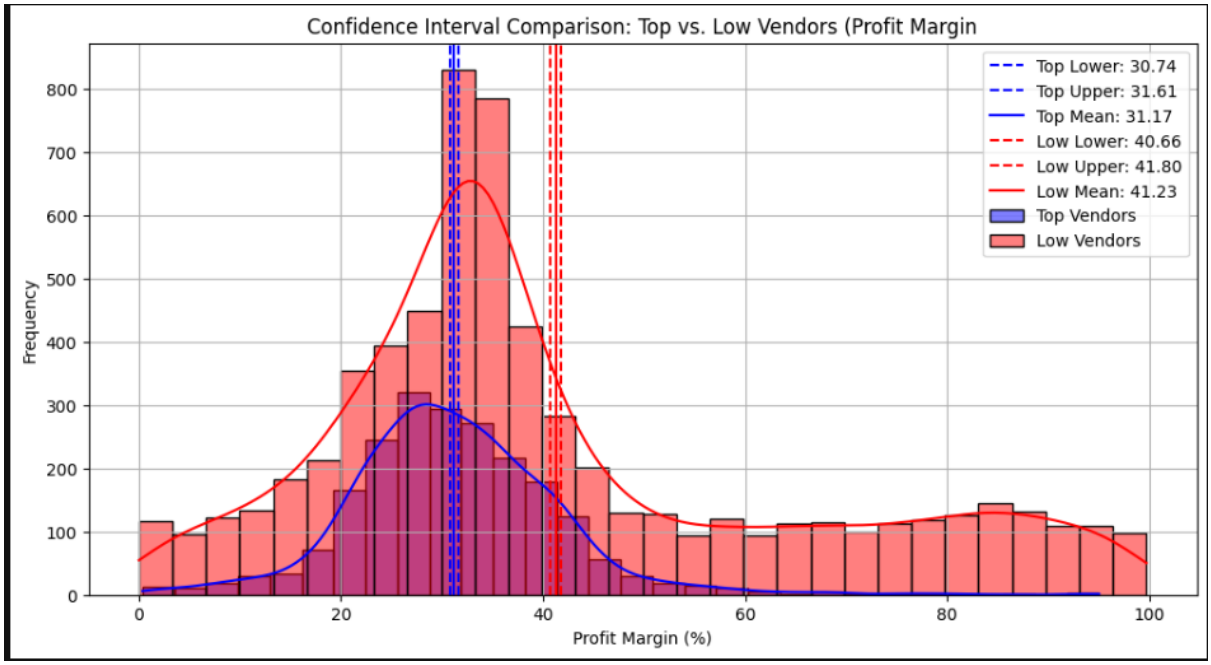
Top Vendors' Profit Margin (95% CI): (30.74%, 31.61%), Mean: 31.17%

Low Vendors' Profit Margin (95% CI): (40.48%, 42.62%), Mean: 41.55%

Low-performing vendors maintain higher margins but struggle with sales volumes, indicating potential pricing inefficiencies or market reach issues.

Actionable Insights:

- Top-performing vendors: Optimize profitability by adjusting pricing, reducing operational costs, or offering bundled promotions.
- Low-performing vendors: Improve marketing efforts, optimize pricing strategies, and enhance distribution networks.



Final Recommendations

Re-evaluate pricing for low-sales, high-margin brands to boost sales volume without sacrificing profitability.

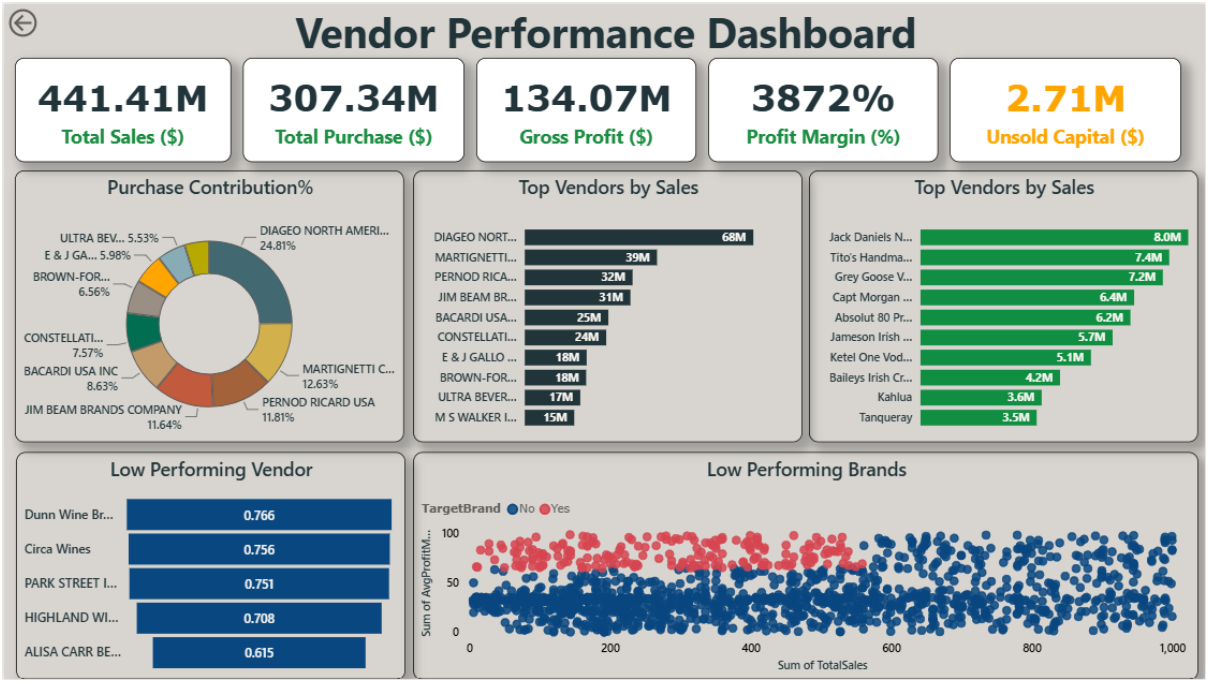
Diversify vendor partnerships to reduce dependency on a few suppliers and mitigate supply chain risks.

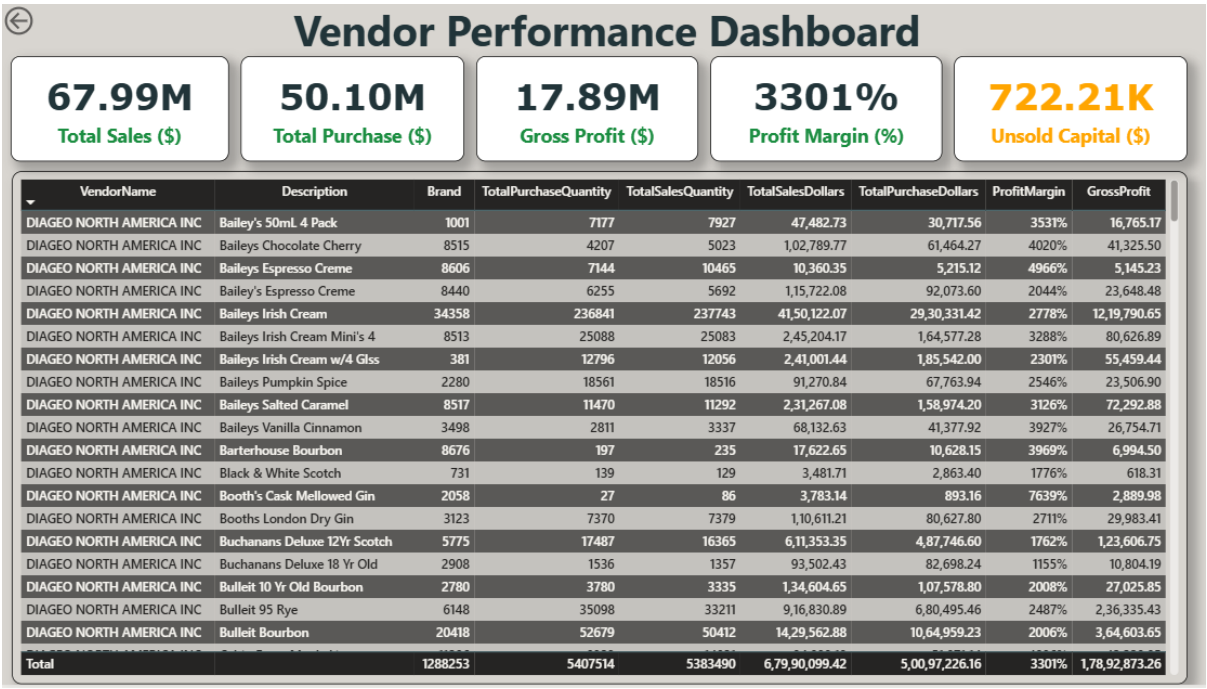
Leverage bulk purchasing advantages to maintain competitive pricing while optimizing inventory management.

Optimize slow-moving inventory by adjusting purchase quantities, launching clearance sales, or revising storage strategies.

Enhance marketing and distribution strategies for low-performing vendors to drive higher sales volumes without compromising profit margins.

By implementing these recommendations, the company can achieve sustainable profitability, mitigate risks, and enhance overall operational efficiency





VendorName	Description	Brand	TotalPurchaseQuantity	TotalSalesQuantity	TotalSalesDollars	TotalPurchaseDollars	ProfitMargin	GrossProfit
DIAGEO NORTH AMERICA INC	Bailey's 50mL 4 Pack	1001	7177	7927	47,482.73	30,717.56	3531%	16,765.17
DIAGEO NORTH AMERICA INC	Baileys Chocolate Cherry	8515	4207	5023	1,02,789.77	61,464.27	4020%	41,325.50
DIAGEO NORTH AMERICA INC	Baileys Espresso Creme	8606	7144	10465	10,360.35	5,215.12	4966%	5,145.23
DIAGEO NORTH AMERICA INC	Bailey's Espresso Creme	8440	6255	5692	1,15,722.08	92,073.60	2044%	23,648.48
DIAGEO NORTH AMERICA INC	Baileys Irish Cream	34358	236841	237743	41,50,122.07	29,30,331.42	2778%	12,19,790.65
DIAGEO NORTH AMERICA INC	Baileys Irish Cream Mini's 4	8513	25088	25083	2,45,204.17	1,64,577.28	3288%	80,626.89
DIAGEO NORTH AMERICA INC	Baileys Irish Cream w/4 Glss	381	12796	12056	2,41,001.44	1,85,542.00	2301%	55,459.44
DIAGEO NORTH AMERICA INC	Baileys Pumpkin Spice	2280	18561	18516	91,270.84	67,763.94	2546%	23,506.90
DIAGEO NORTH AMERICA INC	Baileys Salted Caramel	8517	11470	11292	2,31,267.08	1,58,974.20	3126%	72,292.88
DIAGEO NORTH AMERICA INC	Baileys Vanilla Cinnamon	3498	2811	3337	68,132.63	41,377.92	3927%	26,754.71
DIAGEO NORTH AMERICA INC	Barterhouse Bourbon	8676	197	235	17,622.65	10,628.15	3969%	6,994.50
DIAGEO NORTH AMERICA INC	Black & White Scotch	731	139	129	3,481.71	2,863.40	1776%	618.31
DIAGEO NORTH AMERICA INC	Booth's Cask Mellowed Gin	2058	27	86	3,783.14	893.16	7639%	2,889.98
DIAGEO NORTH AMERICA INC	Booths London Dry Gin	3123	7370	7379	1,10,611.21	80,627.80	2711%	29,983.41
DIAGEO NORTH AMERICA INC	Buchanans Deluxe 12Yr Scotch	5775	17487	16365	6,11,353.35	4,87,746.60	1762%	1,23,606.75
DIAGEO NORTH AMERICA INC	Buchanans Deluxe 18 Yr Old	2908	1536	1357	93,502.43	82,698.24	1155%	10,804.19
DIAGEO NORTH AMERICA INC	Bulleit 10 Yr Old Bourbon	2780	3780	3335	1,34,604.65	1,07,578.80	2008%	27,025.85
DIAGEO NORTH AMERICA INC	Bulleit 95 Rye	6148	35098	33211	9,16,830.89	6,80,495.46	2487%	2,36,335.43
DIAGEO NORTH AMERICA INC	Bulleit Bourbon	20418	52679	50412	14,29,562.88	10,64,959.23	2006%	3,64,603.65
Total		1288253	5407514	5383490	6,79,90,099.42	5,00,97,226.16	3301%	1,78,92,873.26